

“I just see numbers, but how do you feel about your training?”: Clinicians’ Data Needs in Telemonitoring for Colorectal Cancer Surgery Prehabilitation

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ABSTRACT

Prehabilitation for colorectal cancer (CRC) surgery is a multimodal lifestyle change program that prevents postoperative complications. While telemonitoring (TM) during prehabilitation is still in its infancy, this technology is currently being explored to build an understanding of patients’ lifestyles outside the clinic. However, one prevalent design problem is fitting TM systems in clinicians’ existing workflows without disrupting or creating more work. In this study, we investigate how data from a TM system tracking patient physical activity, sleep, and stress fit into the workflow of different prehabilitation clinical specialties. We present early findings from 4 workshops conducted with an oncological physiotherapist, a specialized nurse, a surgeon, and a research doctor interested in policy making. Findings show that: (i) there is a need for different decision-support system designs per specialty, (ii) clinicians’ data needs can change over the program’s timeline, and (iii) clinicians prefer to share the data with patients.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

Patient telemonitoring data; Decision-support systems; Prehabilitation for cancer surgery



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1 INTRODUCTION

Within habilitation programs such as prehabilitation and rehabilitation, advances in telemonitoring (or “remote monitoring”) technologies nowadays allow clinicians to understand the lifestyle of patients outside the clinic to improve clinical decision-making [5, 16, 26]. Telemonitoring (TM) systems transmit self-reported metrics and physiological data collected through mobile or sensor-based devices (such as blood pressure monitors, ECG trackers, motion sensors) directly to care providers via automated electronic means [5]. Although there have been extensive efforts to implement TM systems in habilitation programs in the past decade, these technologies fail to be widely adopted in clinical environments because they disrupt the clinicians’ current workflows and add extensive data work to their load [4, 11, 13, 27]. In this research, we focus on facilitating clinical decision-support in telemonitoring patient behaviors within the context of prehabilitation for colorectal cancer (CRC) surgery.

Prehabilitation for CRC surgery is part of habilitation services, defined as “evidence-based healthcare programs focused on helping patients keep, get back or improve skills for daily living” [25]. Prehabilitation takes place before surgery and aims at increasing functional capacity in anticipation of the upcoming stress of the operation [21]. During the 4 to 6-week program, CRC patients scheduled for abdominal surgery participate in a series of interventions, such as dietary and psychological counseling with a focus on

Table 1: Card topics and contents used in the workshops

Topic and description	Cards
Features - Descriptions of the features that can be derived from the self-reported and sensor data collected with the TM system; each card had example graphs and explanations of the possible derived features on the back for clarity	Heart rate (HR) (with derived features like resting HR, ST-segment analysis, HR recovery); Heart rate variability (HRV) (with derived features like mean HRV, time-domain features, frequency-domain features); Sleep (with derived features like duration, start and end, quality); Accelerometer (with derived features like effort, steps, time spent in activities with different intensities)
Themes - Broader topics inferred from a combination of features; each card had explanations on how the themes can be derived from features, and examples of sensor data graphs and self-reported questions on the back	Stress (sensor and self-reported); Fatigue (sensor and self-reported); Sleep (sensor and self-reported); Activity during prehab training (sensor); Activity outside prehab training (sensor and self-reported); Functional capacity (sensor); Barriers and enablers of physical activity (self-reported)
Analytic tasks - High-level or low-level analytic tasks that people perform when interacting with data, derived from two relevant taxonomies [3, 33]; the tasks were divided into umbrella categories for easier navigation	Have a generic view - Overview, History ; Find a specific value - Identify, Locate, Retrieve value, Compute derived value, Find extremum ; Find connections/differences - Distinguish, Categorize, Compare, Correlate ; Obtain details - Details-on-demand, Zoom, Filter, Extract ; Determine a group of values - Rank, Cluster, Find anomalies, Determine range, Distribution
Stakeholders and population characteristics - Cards describing stakeholders actively involved in prehabilitation and characteristics of the data-generating population	Stakeholders - Doctor, Patient, Nurse, Physiotherapist ; Population - Single person, Group, Age, Gender
Time parameters - Description of different time-related aspects which participants might consider in their interactions with data	When - Minute, Hour, Day, Week, Month, Season, Year ; Time of the day - Morning, Afternoon, Evening, Night ; Normal vs. special time - Weekend, Weekday, Holiday, Day before/after ; Event characteristics - Duration, Frequency, Continuous event, Gaps

supervised physical activity, with the aim of preventing postoperative complications and improving recovery [29]. Prehabilitation is proven to be clinically effective; however, current studies pinpoint barriers such as the lack of personalized exercise recommendations, programs tailored to the patient's individual needs, and supervision in the home environment to stimulate physical activity [2, 16]. To address these barriers, recent work has been investigating the implementation of TM systems in prehabilitation for major surgery to provide clinicians with insights into patient's physical activity and well-being at home [9, 10, 16].

Although TM in rehabilitation programs is a relatively well-investigated topic [4, 13], TM in prehabilitation is a less familiar subject. Existing studies investigate the technical feasibility of specific sensors [9, 10] or the feasibility of using TM in tele-prehabilitation programs [16]. Nonetheless, there is a lack of studies investigating clinicians' needs and expectations in TM for prehabilitation for major surgery. Failing to identify and address these needs and expectations can result in poor system adoption and adherence [35]. Earlier research found that not reflecting prior expectations of clinicians regarding TM in actual experiences could lead to disappointment [12]. Additional work on clinical perceptions of TM emphasizes that quite often, clinicians' disappointment is led by the realization that managing and interpreting patient-generated data does not reduce their workload but requires re-adapting current workflows [11, 17, 20, 35]. This can impede the user's perception of the beneficial effects of the system and, ultimately, affect the success of the TM systems [4]. In this study, we address this gap in knowledge related to clinician expectations and needs regarding 'data work' (i.e., managing, analyzing, and interpreting data) in TM during prehabilitation for CRC surgery.

Our research builds on previous CSCW literature investigating implementation opportunities and barriers of TM systems in chronic disease management [22], technology implementation in prehabilitation [37], as well as clinician perspectives on data work in healthcare [8, 27]. We use data as human-centric design material to obtain insights into preferred data practices in TM for prehabilitation for CRC surgery that could eventually inform and support the design of such systems [18]. We use cards depicting patient-generated data from a TM system tracking patient physical

activity, sleep, and stress to understand preferred workflows of various clinical specialties of a prehabilitation multi-disciplinary team (MDT) from a teaching hospital in the Netherlands. Our work is driven by the following questions: (i) *How does the data collected through a TM system fit into the current workflows of MDT clinicians given their various workflows?* and (ii) *What analytical decisions do clinicians need to make when working with TM data?* We present preliminary results from four workshops with four stakeholders having different specialties related to the prehabilitation ecosystem - an oncology physiotherapist, a specialized nurse, a research clinician specialized in CRC prehabilitation, and a surgeon.

2 METHODOLOGY

For this study, we collaborate with clinicians from a teaching hospital in the Netherlands, where the current state-of-the-art TM system for prehabilitation consists of an on-body wearable that collects electrocardiogram (ECG) data, a wristband that collects raw acceleration (ACC) data, and a mobile application that collects daily self-reported data describing the patient's activities and well-being (e.g., sleep quality, motivation for physical activity, stress). This system focuses on gathering data about functional capacity, physical activity, sleep, and stress¹. We use this system as a starting point for discussions around the types and characteristics of TM data that could be useful for different clinicians in the context of prehabilitation.

We follow a qualitative study design² based on a combination of tutorial interviews [23] and card-based workshops. Card-based tool-kits are effective in triggering idea generation and the exploration of abstract concepts in the design process [19, 30, 32, 36]. In card-sorting methods [14, 31], cards are used to understand the mental models that users form about concepts [1]. Our participants' backgrounds and data literacy levels are diverse. Therefore, understanding and remembering all collected data types during an interview without any visual aids might have been difficult and hindered envisioning potential application scenarios. Inspired by

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²The study was approved by the Ethical Review Board of our university (ref no. ERB2022ID104)

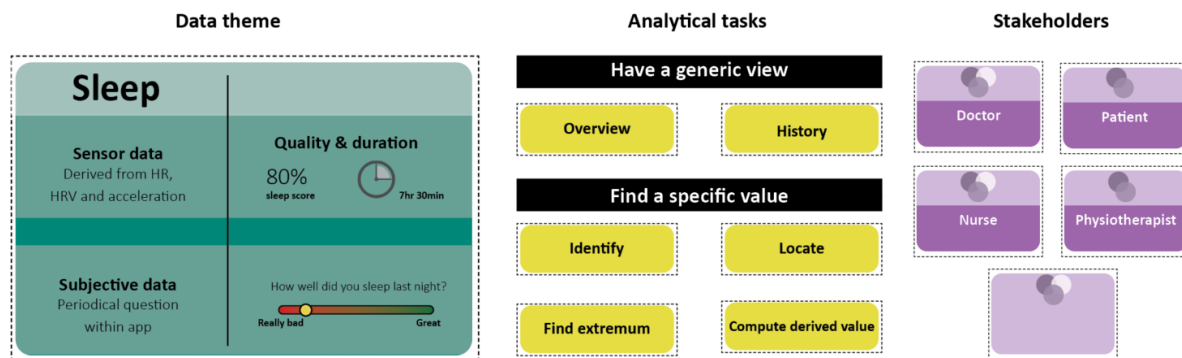


Figure 1: Pre-print examples of cards: (left) data theme with explanations on the inside about sensor as well as self-reported data describing the theme; (middle) categories of analytical tasks under umbrella themes; (right) stakeholder cards with example empty card participants can fill in

card-sorting methods, we developed our card-based workshop to address these potential obstacles.

Based on a stakeholder mapping session conducted using the IGOHcaps method [24], we identified the stakeholders that play a key role in the CRC prehabilitation pathway and could benefit most from interacting with the TM data. Out of this group, we recruited four clinicians so far: **an oncological physiotherapist (P1), a specialized nurse (P2), a research doctor interested in policy making (P3), and a surgeon (P4)**. Each workshop session lasted 60 to 80 minutes and included two researchers, one clinician, and a translator when needed. Participants are native-Dutch speakers and English is their second language. Quotes are reported verbatim and might include grammar and spelling mistakes. The structure of the workshops is as follows: (i) **Tutorial interview** - the participant is asked to describe their current workflow related to the prehabilitation pathway, highlighting the moments that include interactions with any types of patient data or information; (ii) **Card-based workshop** - the participant interacts with cards related to the measured data in the context of their workflow, elaborating on their choices (Figure 2). At the start of the activity, participants are given an empty template of a user journey, writing materials, and sticky notes to draw their current workflow collaboratively with the researchers. Following, cards are used as prompts to generate and express ideas related to TM patient data [30, 32]. Multiple instances of each card were printed and stacked on A3 templates using putty-like reusable adhesive to allow for interactivity. Each template contained cards related to specific topics (Table 1) as well as empty cards, allowing clinicians to add missing topics of interest. The sessions were video recorded for post hoc analysis after obtaining consent from the participants.

The templates were introduced sequentially. First, the *Themes* and *Features* templates were introduced. Participants were asked to describe which cards they found interesting or relevant to their work. The other templates were used to further elaborate on these themes - for example, cards related to *Analytic tasks* were used to describe how the participants would like to analyze this data, while the *Stakeholders* cards were used to elaborate on who else could be included in their interactions with the data. Participants were also asked to elaborate on how the selected cards would fit into their workflow and asked to include them in the user journey

constructed in the first activity. We analyzed interview data using axial coding [34], focusing primarily on emerging themes related to their workflow and data needs, and secondarily on the effects of the materials on idea generation and expression. To enhance the trustworthiness of the process [28], two authors generated codes separately while reading the interview transcripts. Then, they compared the codes and went through multiple iterations of re-coding and identifying connections between the codes, until agreement was established within the research team.

3 INITIAL INSIGHTS AND TAKEAWAYS

Preferred data for clinical decision-making - In line with previous research stressing the difficulty to integrate TM data in clinicians' busy workflows [15, 22, 35], participants emphasize that dashboards for TM systems should only highlight "essential information". Our findings underline, however, the challenge of aligning the data needs of various stakeholders so that they can all benefit from the same decision-support system. As physical activity is the main component of prehabilitation programs, overseeing activity patterns and intensity (during or outside training sessions), as well as observing functional capacity trends over time were considered important by all our participants. However, their interest in other data types varied based on medical specialty and moments of interaction with the patient. All presented data types were considered essential for the surgeon and the research clinician, for improving the overall program (based on general population data sets) and for personalization (based on individual data). The other clinicians were more interested in data about physical activity paired with barriers and facilitators of activity, information that is essential to nurses and physiotherapists to establish a clear patient profile before offering lifestyle advice. TM data could complement the current process of openly discussing this information with the patient, as patients might be uncertain or ashamed to report the truth in these discussions. Fatigue and sleep data are also useful for the physiotherapist, to adjust the training intensity before each session. Nevertheless, more subjective, self-reported data, such as drinking and smoking habits, as well as dietary intake and social support, not captured by the current system, can help the nurse understand certain behavioral patterns and identify alarming situations. These

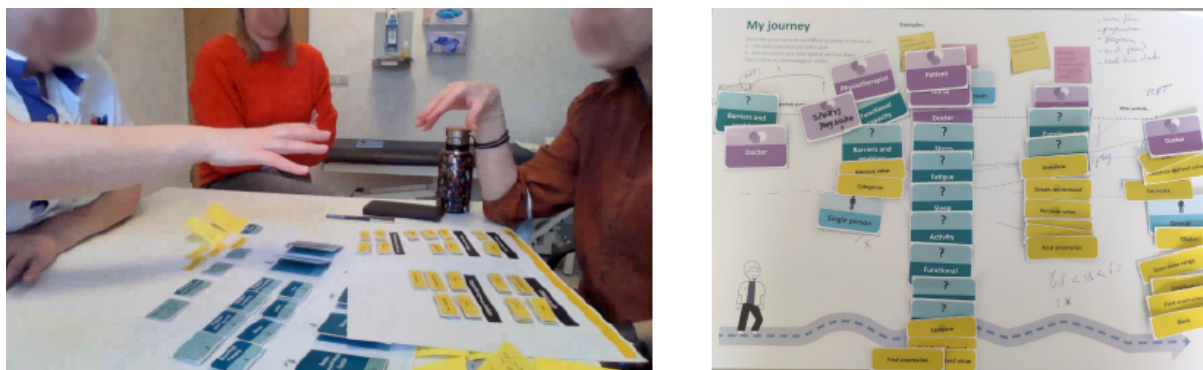


Figure 2: (left) P2 opening data cards for more information about collected data; (right) Outcome of the card-sorting session: P4 integrating all cards in the current workflow they described.

findings complement previous studies on perspectives of clinicians that would benefit from interacting with patient data [7, 8, 22] by mapping the diversity in what each clinician wants to see in the data. Although TM data offers useful insights, the exposure to all this data might be overwhelming when there is very limited time to interact with it, and unnecessary based on the clinicians' data needs. Decision-support systems based on TM data need to offer interface customization based on different clinical profiles (e.g., nurse vs. physiotherapist) so that it allows each specialty to view only the most essential information and perform the types of analysis they are interested in.

Analytical tasks for corresponding data needs – Regarding analytical tasks performed while interacting with the data, practicing clinicians considered having a high-level overview per patient the most important when dealing with heavy workloads. Participants mentioned that data overviews need to be always available, nonetheless, only checked if needed (e.g., if a patient contacts them with a complaint). Determining the range of functional capacity during the program was deemed an important task, as the main focus of prehabilitation is to increase the patient's functional capacity. Nonetheless, clinicians know little about a patient's progression of functional capacity over the course of the program at the moment - TM systems can, therefore, help provide these insights. Participants wished to identify anomalies in stress, sleep, and fatigue for MDT discussions about individual patient cases or in-the-moment decision-making (e.g., in the case of an urgent situation). Anomalies, however, need to be obviously signaled (e.g., through notifications or color changes), as participants lack the time to periodically check data reports. Identifying correlations between different data (such as sleep, stress, and physical activity) was preferred when investigating why certain anomalies might occur. Practicing clinicians were not interested in complex analytical tasks due to the personal nature of their work, and the lack of time they have to sit in front of their screens. However, clinicians concerned with policymaking identified tasks such as "computing derived value" of physical activity and "finding extremum" of fatigue or sleep data as important for optimizing program policies. Therefore, systems using TM patient data should encompass both generic data visualization reports as well as more detailed data analysis functionalities for each supported clinical profile.

Matching data needs to program timeline – Results highlight that different types of data are preferred based on how valuable the information is at different points in the prehabilitation timeline. Participants agree that barriers and enablers of physical activity are important at the beginning of the program during the MDT meeting, at which point they can determine the details of the program and adapt it if needed. An overview of functional capacity progression is needed 3-4 days before surgery to determine if the surgery should be postponed or canceled in case of extreme frailty. Conversely, an indication of perceived stress is most valuable at the patient's first consultation with the surgeon, since the healthcare professional can help alleviate the patient's anxieties at that moment - *"The case manager [i.e., nurse] does a lot, but they [i.e., the patient] want to talk to the doctor. And I think that one of my major goals in the outpatient clinic is to calm people down"* (P4). In-the-moment decision-making is preferred for practicing clinicians (P1,2 and 4) due to limited time. In these cases, functional capacity, stress, and sleep data would be reviewed during consultations, but only if there are anomalies (e.g., unusual sleep or stress patterns, or a sudden increase in fatigue) or if the patient mentions any issues - *"I have no time to see a lot of data which is good or [has] no problems."* (P2). Data are also valuable after each program finishes when viewed at a cohort level to optimize the program (P3, P4). To our knowledge, the varying significance of different data types during the prehabilitation journey has not been discussed yet in previous research. Designers of decision-support TM systems need to be mindful that clinicians' data needs are dynamic over time, and requirements regarding their needs should be elicited based on the program's timeline.

Sharing data with other stakeholders – Our findings align with the results of previous studies emphasizing the potential role that data and information technologies can play in supporting patient-doctor communication [8, 20]. Participants highlight the value data can have as an object for discussion which can enhance the connection they already develop with the patients. In-the-moment discussions about alarming anomalies and data overviews during consultations with the patient are preferred for multiple reasons: a) to understand the patient’s point of view - *“I would just ask them about it: ‘I just see numbers, but how do you feel about your training?’”* (P4), b) to avoid de-personalization of data - *“that’s really useful for us - [...] if you get more information that makes you understand this person [i.e., the patient] and talk to them better*

and to help them.” (P1) and c) to give patients control over their patient journey - “I can tell them: ‘You’re 70 years old, you are a frail patient, but see what you did in 2 weeks! Keep on going!’ [...] They feel involved, there’s a lot of attention being paid to them, they don’t just feel like a number.” (P4). Nonetheless, the ecosystem of people who experience the data also expands to include other clinicians during the MDT meetings before and after the surgery or the patient’s partner for a better understanding of their everyday contexts. Our preliminary results suggest that data can be co-experienced, as meaning is created through collaboratively interacting with the data [6]. Hence, decision-support systems should acknowledge and encourage communication around data between multiple stakeholders. However, co-experiencing data adds a layer of complexity - systems should, for example, accommodate data representations for both high as well as low data literacy (e.g., the data selected by the clinician, for example, might not be aligned with the questions and interests of the patient [8]).

4 CONCLUSION AND FUTURE WORK

Although patient TM systems are increasingly used in prehabilitation programs, there is a lack of knowledge on how these systems can fit into clinicians’ workflows in prehabilitation for CRC surgery. While our findings only reflect the experiences and opinions of a small sample of clinicians working at the same clinic, they highlight the difficulty of aligning the diverse data needs in the same decision-support system. Dashboards for TM systems should cater to the data needs of various clinical specialties, account for needs that dynamically change over the program’s timeline, and enable data co-experience between clinicians, patients, and other stakeholders. Our future work will focus on investigating the needs of other specialties of the MDT, as well as looking into how resulting TM system guidelines can apply to other habilitation programs such as rehabilitation.

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